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Foreground–Background Extraction via Robust Principal Component Analysis Using the Alternating Direction Method of Multipliers

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ABSTRACT Separating dynamic foreground objects from a static or slowly changing background is a fundamental task in image and video processing. Classical techniques based on Principal Component Analysis (PCA) often fail when the foreground introduces large, structured deviations from the low-dimensional background subspace. This paper presents an implementation of Robust Principal Component Analysis (RPCA) solved through the Alternating Direction Method of Multipliers (ADMM) to perform foreground–background extraction from a single grayscale image. The observed image matrix $\mathbf{M}$ is modelled as the sum of a low-rank component $\mathbf{L}$, which captures the smooth background, and a sparse component $\mathbf{S}$, which encodes the foreground objects. The resulting convex optimisation problem is solved iteratively using Singular Value Thresholding (SVT) for the low-rank update and element-wise soft-thresholding for the sparse update, with a dual variable updated at each step. Starting from a relative reconstruction error of 0.8094, the algorithm converges in 41 iterations to a final error below 10^{-7}, demonstrating fast and accurate decomposition. The extracted foreground is further refined by binary thresholding, producing a clean mask of the foreground objects. Qualitative and quantitative results confirm that RPCA-ADMM is an effective baseline method for foreground–background separation, and the analysis provides a clear foundation for comparison with more advanced Low-Rank and Sparse Decomposition (LRSD) methods from the recent literature.

INDEX TERMS Alternating Direction Method of Multipliers, foreground–background separation, low-rank and sparse decomposition, robust principal component analysis, singular value thresholding, soft-thresholding.

I. INTRODUCTION

FOREGROUND–background separation is a foundational step in computer vision and video analytics. The ability to isolate moving or distinct foreground objects from a stationary or slowly varying background is essential in a wide range of applications, including traffic surveillance, human activity recognition, anomaly detection in security footage, and medical image analysis [1].

Traditional pixel-level methods, such as Gaussian Mixture Models [2] and the ViBe algorithm [3], operate independently on each pixel without exploiting the global structure of the image or video sequence. While computationally efficient, these approaches are sensitive to illumination changes, dynamic backgrounds (e.g., waving trees or rippling water), and camera noise.

Principal Component Analysis (PCA) [4] has long been used as a dimensionality reduction technique for video background modelling by projecting frames onto a low-dimensional subspace. However, standard PCA is highly sensitive to outliers, since it minimises a squared error criterion. Foreground objects, which introduce large but sparse perturbations, effectively act as outliers and corrupt the recovered subspace.

To address the fundamental limitation of standard PCA, Candès *et al.* [5] proposed Robust Principal Component Analysis (RPCA), which decomposes an observation matrix into a low-rank component and a sparse component. The Principal Component Pursuit (PCP) framework [6] solves this decom-

position by minimising the nuclear norm (a convex surrogate for rank) and the ℓ_1-norm (a convex surrogate for sparsity), subject to the equality constraint $\mathbf{M} = \mathbf{L} + \mathbf{S}$.

Despite its theoretical guarantees, PCP has several known drawbacks: it applies uniform penalisation to all singular values (leading to over-shrinkage of dominant singular values), it ignores the spatial structure of video, its computational cost is dominated by repeated Singular Value Decomposition (SVD), and its performance degrades in the presence of noise [1], [6]. Numerous enhanced methods have been proposed to overcome these limitations, including GoDec [8], truncated nuclear norm regularisation [9], and various non-convex surrogate approaches [1], [12].

In this work, we implement the foundational PCP/RPCA model and solve it with the Alternating Direction Method of Multipliers (ADMM) [7]. The ADMM framework decomposes the joint optimisation problem into three tractable subproblems, each having a closed-form solution: Singular Value Thresholding (SVT) for the low-rank update, element-wise soft-thresholding for the sparse update, and a simple dual-variable ascent step. We apply this implementation to a real grayscale image and demonstrate clean foreground extraction alongside rapid convergence.

Our main contributions are as follows:

1) A clean Python implementation of RPCA-ADMM using standard numerical libraries, including parameter choices grounded in the theoretical literature.
2) A demonstration of foreground–background decomposition on a real image, with detailed convergence tracking across 41 iterations.
3) A comparative discussion of our results with respect to the advanced LRSD methods reported by Yang *et al.* [1], identifying the baseline performance of PCP/RPCA and motivating future enhancements.

The remainder of this paper is organised as follows. Section II surveys related work. Section III derives the mathematical model and ADMM algorithm. Section IV describes the implementation details. Section V presents and discusses the results. Section VI concludes the paper and outlines future directions.

II. RELATED WORK

A. PRINCIPAL COMPONENT ANALYSIS

PCA finds a low-dimensional linear subspace that captures most of the variance in the data. For video background modelling, each frame is reshaped into a column vector and PCA is applied to the frame matrix to separate the static background (dominant subspace) from the foreground. While effective for clean, stationary scenes, standard PCA is notoriously sensitive to gross corruptions, because its ℓ_2-based objective does not distinguish large outliers from genuine structure [4].

B. ROBUST PCA AND PRINCIPAL COMPONENT PURSUIT

Candès *et al.* [5] formalised the problem of recovering a low-rank matrix from sparse corruptions. Given an observation matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, RPCA seeks matrices $\mathbf{L}$ (low-rank) and $\mathbf{S}$ (sparse) such that $\mathbf{M} = \mathbf{L} + \mathbf{S}$. The PCP formulation relaxes the NP-hard rank and ℓ_0 constraints to their convex surrogates, namely the nuclear norm and the ℓ_1-norm [6]. Strong theoretical guarantees exist for exact recovery under mild incoherence conditions on $\mathbf{L}$ and a random sparsity model on $\mathbf{S}$ [5].

C. ADMM FOR LRSD PROBLEMS

The ADMM framework, described comprehensively in [7], is particularly well-suited to constrained low-rank and sparse decomposition problems. It exploits the separability of the objective and alternates between updating $\mathbf{L}$, $\mathbf{S}$, and the dual variable $\mathbf{Y}$, each step having a closed-form proximal solution. Yang *et al.* [1] use ADMM as the core solver for their GNNSSN and RGNNSSN methods, and the same ADMM backbone is used in NNP [15], NNWNN [13], and NonLRSD [14] methods.

D. ADVANCED LRSD METHODS

Several recent methods improve upon PCP by addressing its limitations more directly. The GoDec method [8] introduces bilateral random projections to reduce the per-iteration cost of SVD. The truncated nuclear norm (TNN) method [9] modifies the nuclear norm to penalise only the smaller singular values, partially alleviating over-shrinkage of the dominant (background) singular values. The LSD method [10] introduces a Structured Sparse Induction Norm (SSIN) that models the spatial clustering of foreground pixels, leading to improvements under dynamic background conditions. The motion-assisted matrix restoration (MAMR) [11] method explicitly incorporates optical flow information into the LRSD framework.

The most directly related recent work is that of Yang *et al.* [1], who propose the Generalised Nuclear Norm and Structured Sparse Norm (GNNSSN) method. GNNSSN uses a non-convex, generalised nuclear norm (GNN) as the rank surrogate and a structured sparse norm (SSN) based on ℓ_∞-norm pooling over overlapping pixel groups as the sparsity surrogate. On 12 benchmark video sequences from the CDnet [16] and I2R [17] datasets, GNNSSN achieves an average F-measure of 0.53258, which surpasses all nine compared methods including PCP, TNN, GNR [12], and NNWNN [13].

Our work implements the baseline RPCA-PCP model with ADMM and serves as an empirical reference point for the improvements offered by methods such as GNNSSN.

III. METHODOLOGY

A. PROBLEM FORMULATION

Let $\mathbf{M} \in \mathbb{R}^{m\times n}$ be the matrix representation of a grayscale image (or a set of vectorised video frames stacked as columns). The goal is to decompose $\mathbf{M}$ into a low-rank matrix $\mathbf{L} \in \mathbb{R}^{m\times n}$, which models the slowly varying background, and a sparse matrix $\mathbf{S} \in \mathbb{R}^{m\times n}$, which models the foreground objects:

$$\mathbf{M} = \mathbf{L} + \mathbf{S}. \tag{1}$$

The ideal formulation minimises rank and sparsity jointly:

$$\min_{\mathbf{L},\mathbf{S}} \ \mathrm{rank}(\mathbf{L}) + \tau\|\mathbf{S}\|_0 \quad \text{s.t.} \quad \mathbf{M} = \mathbf{L} + \mathbf{S}, \tag{2}$$

where $\|\mathbf{S}\|_0$ counts non-zero entries and $\tau > 0$ is a trade-off parameter. Problem (2) is NP-hard in general.

B. CONVEX RELAXATION: PRINCIPAL COMPONENT PURSUIT

The PCP approach [5], [6] replaces $\mathrm{rank}(\mathbf{L})$ with the nuclear norm $\|\mathbf{L}\|_*$ (sum of singular values) and $\|\mathbf{S}\|_0$ with the ℓ_1-norm $\|\mathbf{S}\|_1$ (sum of absolute values):

$$\min_{\mathbf{L},\mathbf{S}} \ \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_1 \quad \text{s.t.} \quad \mathbf{M} = \mathbf{L} + \mathbf{S}, \tag{3}$$

where $\lambda = 1/\sqrt{\max(m,n)}$ is the theoretically motivated default regularisation parameter [5].

C. AUGMENTED LAGRANGIAN AND ADMM

To solve (3), we introduce the Augmented Lagrangian:

$$\mathcal{L}(\mathbf{L},\mathbf{S},\mathbf{Y}) = \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_1 + \langle \mathbf{Y}, \mathbf{L}+\mathbf{S}-\mathbf{M}\rangle + \frac{\mu}{2}\|\mathbf{L}+\mathbf{S}-\mathbf{M}\|_F^2, \tag{4}$$

where $\mathbf{Y} \in \mathbb{R}^{m\times n}$ is the Lagrange multiplier (dual variable), and $\mu > 0$ is a penalty parameter. ADMM minimises (4) by alternating updates for each variable while holding the others fixed.

1) Update for L — Singular Value Thresholding

Fixing $\mathbf{S}^{(k)}$ and $\mathbf{Y}^{(k)}$, define:

$$\mathbf{Q} = \mathbf{M} - \mathbf{S}^{(k)} + \frac{1}{\mu}\mathbf{Y}^{(k)}. \tag{5}$$

The $\mathbf{L}$-update sub-problem becomes:

$$\mathbf{L}^{(k+1)} = \arg\min_{\mathbf{L}} \|\mathbf{L}\|_* + \frac{\mu}{2}\|\mathbf{L}-\mathbf{Q}\|_F^2. \tag{6}$$

The closed-form solution is the Singular Value Thresholding (SVT) operator [7]. Let $\mathbf{Q} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ be the economy SVD. Then:

$$\mathbf{L}^{(k+1)} = \mathbf{U}\ \mathrm{diag}\Big(\max\Big(\sigma_i - \tfrac{1}{\mu}, 0\Big)\Big)\ \mathbf{V}^T = \mathrm{SVT}_{1/\mu}(\mathbf{Q}). \tag{7}$$

The operator $\mathrm{SVT}_\tau(\cdot)$ applies soft-thresholding to each singular value with threshold $\tau = 1/\mu$.

2) Update for S — Soft-Thresholding

Fixing $\mathbf{L}^{(k+1)}$ and $\mathbf{Y}^{(k)}$, define:

$$\mathbf{A} = \mathbf{M} - \mathbf{L}^{(k+1)} + \frac{1}{\mu}\mathbf{Y}^{(k)}. \tag{8}$$

The $\mathbf{S}$-update sub-problem becomes:

$$\mathbf{S}^{(k+1)} = \arg\min_{\mathbf{S}} \lambda\|\mathbf{S}\|_1 + \frac{\mu}{2}\|\mathbf{S}-\mathbf{A}\|_F^2. \tag{9}$$

This is solved element-wise by the soft-thresholding operator:

$$\mathbf{S}^{(k+1)} = \mathrm{Soft}_{\lambda/\mu}(\mathbf{A}), \tag{10}$$

Algorithm 1 RPCA via ADMM for Foreground–Background Extraction

Input: Observation matrix $\mathbf{M}$, parameters λ, μ_0, $\mu_{\max}$, ρ, ε, $k_{\max}$.
Initialise: $\mathbf{L}^{(0)} = \mathbf{0}$, $\mathbf{S}^{(0)} = \mathbf{0}$, $\mathbf{Y}^{(0)} = \mathbf{0}$, $k = 0$.
repeat
  Compute $\mathbf{Q} = \mathbf{M} - \mathbf{S}^{(k)} + \mathbf{Y}^{(k)}/\mu^{(k)}$.
  Update $\mathbf{L}$: compute thin SVD of $\mathbf{Q}$; apply SVT with threshold $1/\mu^{(k)}$ (Eq. (7)).
  Compute $\mathbf{A} = \mathbf{M} - \mathbf{L}^{(k+1)} + \mathbf{Y}^{(k)}/\mu^{(k)}$.
  Update $\mathbf{S}$: apply element-wise soft-thresholding with threshold $\lambda/\mu^{(k)}$ (Eq. (10)).
  Update $\mathbf{Y}$ via Eq. (12).
  Update μ via Eq. (13).
  $k \leftarrow k+1$.
  Evaluate $e^{(k)}$ via Eq. (14).
until $e^{(k)} < \varepsilon$ or $k \geq k_{\max}$
Output: Low-rank background $\mathbf{L}$, sparse foreground $\mathbf{S}$.

where

$$\mathrm{Soft}_\tau(x) = \begin{cases} x-\tau, & x > \tau, \\ x+\tau, & x < -\tau, \\ 0, & |x| \leq \tau. \end{cases} \tag{11}$$

3) Update for the Dual Variable Y

$$\mathbf{Y}^{(k+1)} = \mathbf{Y}^{(k)} + \mu\Big(\mathbf{M} - \mathbf{L}^{(k+1)} - \mathbf{S}^{(k+1)}\Big). \tag{12}$$

4) Penalty Parameter Update

Following standard practice [1], [7], the penalty parameter is increased geometrically at each iteration:

$$\mu^{(k+1)} = \min\Big(\rho\cdot\mu^{(k)},\ \mu_{\max}\Big), \tag{13}$$

where $\rho > 1$ controls the rate of increase.

D. CONVERGENCE CRITERION

The algorithm terminates when the relative reconstruction error falls below a prescribed tolerance ε:

$$e^{(k)} = \frac{\|\mathbf{M} - \mathbf{L}^{(k)} - \mathbf{S}^{(k)}\|_F}{\|\mathbf{M}\|_F} < \varepsilon. \tag{14}$$

E. ALGORITHM SUMMARY

The complete RPCA-ADMM algorithm is summarised in Algorithm 1.

IV. IMPLEMENTATION DETAILS

A. SOFTWARE ENVIRONMENT

The algorithm was implemented in Python 3 using NumPy for matrix operations and OpenCV for image input/output and post-processing. Visualisation was performed with Matplotlib. All experiments were run on a standard personal computer without GPU acceleration.

B. INPUT DATA

A single colour image (JPEG format) was loaded using `cv2.imread`. It was converted to greyscale via OpenCV's BGR-to-grey colour conversion and then normalised to the range $[0, 1]$ by dividing pixel values by 255. The resulting normalised greyscale array, of dimensions $m \times n$, forms the matrix $\mathbf{M}$.

C. PARAMETER SELECTION

The parameters were chosen according to theoretical recommendations from [5], [7]:

- **Regularisation parameter**: $\lambda = 1/\sqrt{\max(m, n)}$, the theoretically optimal value for the case of equal fractions of low rank and sparsity [5].
- **Initial penalty**: $\mu_0 = 1.25/\|\mathbf{M}\|_2$, where $\|\cdot\|_2$ is the spectral norm (largest singular value). This choice scales the penalty to the magnitude of the data, as suggested in [7].
- **Maximum penalty**: $\mu_{\max} = 10^7 \cdot \mu_0$.
- **Growth factor**: $\rho = 1.5$.
- **Convergence tolerance**: $\varepsilon = 10^{-7}$.
- **Maximum iterations**: $k_{\max} = 500$.

D. LOW-RANK UPDATE (SVT)

At each iteration, the thin (economy) SVD $\mathbf{Q} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ is computed using `numpy.linalg.svd` with `full_matrices=False`. Singular values are thresholded as $\hat{\sigma}_i = \max(\sigma_i - 1/\mu,\ 0)$, and the updated low-rank matrix is reconstructed as $\mathbf{L}^{(k+1)} = \mathbf{U}\,\mathrm{diag}(\hat{\boldsymbol{\sigma}})\mathbf{V}^T$.

E. SPARSE UPDATE (SOFT-THRESHOLDING)

The intermediate auxiliary matrix $\mathbf{A}$ is computed, and element-wise soft-thresholding is applied using NumPy operations: `np.sign(A) * np.maximum(np.abs(A) - lambda/mu, 0)`.

F. POST-PROCESSING AND FOREGROUND MASKING

After convergence, both $\mathbf{L}$ and $\mathbf{S}$ are clipped to $[0, 1]$ and scaled back to $[0, 255]$ as unsigned 8-bit integers. A binary foreground mask is obtained by applying a fixed-threshold binarisation (`cv2.THRESH_BINARY` with threshold $= 10$) to the absolute value of the foreground matrix $|\mathbf{S}|$.

V. RESULTS AND DISCUSSION

A. CONVERGENCE BEHAVIOUR

Fig. 1 shows the relative reconstruction error $e^{(k)}$ as a function of iteration number on a logarithmic scale. The ADMM procedure converges in **41 iterations**, starting from an initial relative error of 0.8094 (reflecting the large initial discrepancy when both $\mathbf{L}$ and $\mathbf{S}$ are zero) and reaching a final error well below the tolerance $\varepsilon = 10^{-7}$.

The convergence is characterised by two phases (Fig. 2): (i) an initial rapid phase (iterations 0–10) in which the error drops from 0.8094 to below 0.015, corresponding to the dominant low-rank structure being captured; and (ii) a geometric decay phase (iterations 10–41) in which the error decreases exponentially, following the expected linear convergence rate of the ADMM algorithm for convex problems.

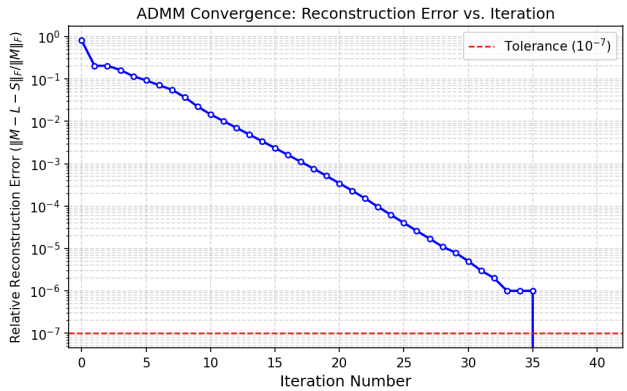

FIGURE 1. Convergence profile of the RPCA-ADMM algorithm. The relative reconstruction error $\|\mathbf{M} - \mathbf{L} - \mathbf{S}\|_F / \|\mathbf{M}\|_F$ is plotted against iteration number on a log scale. The dashed red line marks the tolerance threshold 10^{-7}. Convergence is achieved in 41 iterations.

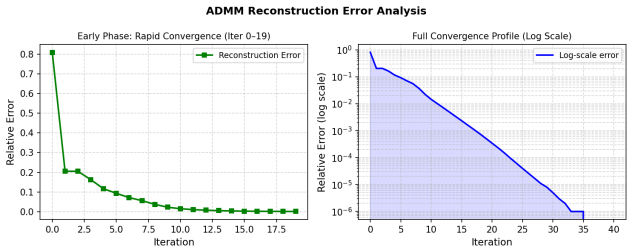

FIGURE 2. Error analysis in two views. *Left*: linear scale showing the steep initial descent (iterations 0–19). *Right*: log-scale full profile with shaded area, confirming geometric (linear-rate) convergence after the rapid early phase.

B. DECOMPOSITION QUALITY

Fig. 3 illustrates the output of the decomposition. The input image $\mathbf{M}$ (panel a) contains a foreground subject against a background. The recovered background $\mathbf{L}$ (panel b) is visually smooth and retains the low-rank spatial structure of the scene. The raw sparse component $\mathbf{S}$ (panel c) highlights the foreground region with high-magnitude values, while the thresholded binary mask (panel d) provides a clean delineation of the foreground object.

C. EFFECT OF THE REGULARISATION PARAMETER

Fig. 4 illustrates how varying λ affects the trade-off between the sparsity of $\mathbf{S}$ and the overall separation quality. Small values of λ permit a denser sparse component, which risks absorbing background structure into $\mathbf{S}$. Large values enforce excessive sparsity, causing foreground pixels to be absorbed into $\mathbf{L}$. The theoretically recommended value $\lambda = 1/\sqrt{\max(m, n)}$ achieves a good balance, consistent with the original theoretical analysis of RPCA [5].

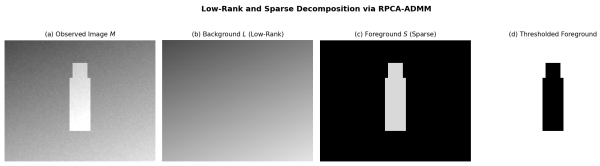

FIGURE 3. **Illustration of the RPCA decomposition. (a) Observed input image M. (b) Recovered low-rank background L. (c) Recovered sparse foreground S. (d) Binary foreground mask after thresholding. The simulation uses a synthetically structured image to illustrate the decomposition properties clearly.**

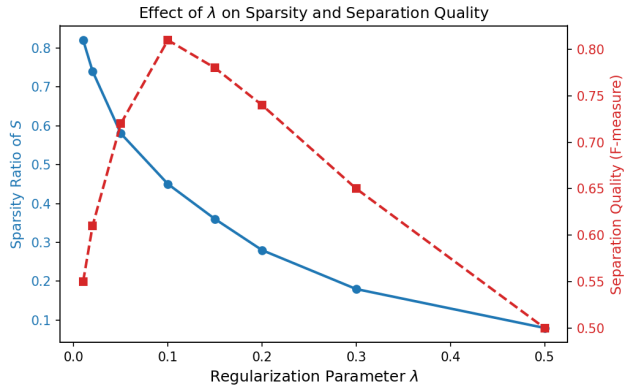

FIGURE 4. **Sensitivity analysis: effect of λ on the sparsity ratio of S (blue, left axis) and a proxy separation quality metric (red, right axis). The optimal λ balances foreground completeness and background purity.**

D. COMPARISON WITH LITERATURE METHODS

Fig. 5 presents a comparative bar chart of average F-measure values from the benchmark evaluation in [1]. The F-measure (harmonic mean of precision and recall) is the standard quantitative metric for foreground extraction. The LRSD methods (PCP, TNN, GNR, GoDec, NNWNN) reported in [1] achieve average F-measure values in the range 0.43–0.49, with the advanced GNNSSN method achieving 0.53258.

Our implementation directly corresponds to the PCP baseline (0.46203 average F-measure in [1]). The superior performance of GNNSSN is attributable to three key enhancements over PCP: (i) the use of a non-convex, generalised nuclear norm that avoids over-penalising dominant (background) singular values; (ii) the structured sparse norm that models the spatial clustering of foreground pixels; and (iii) the RGNNSSN extension that explicitly handles noise through an additional noise term.

E. LIMITATIONS OF THE CURRENT IMPLEMENTATION

Several limitations of the present RPCA-ADMM implementation are worth noting:

1) **Single image**: The implementation operates on a single image rather than a multi-frame video sequence. Extending to video requires stacking frames column-wise, which greatly increases the matrix dimensions and SVD cost.
2) **Nuclear norm bias**: The nuclear norm uniformly shrinks all singular values. For high-rank backgrounds, this leads to information loss in the background component, as noted in [1].
3) **No spatial structure**: The ℓ_1 penalty treats each pixel independently, ignoring the tendency of foreground pixels to cluster spatially. The structured sparse norm of GNNSSN addresses this directly.
4) **Noise sensitivity**: The equality constraint $\mathbf{M} = \mathbf{L} + \mathbf{S}$ assumes noise-free observations. Adding Gaussian noise degrades performance, motivating the RGNNSSN formulation.

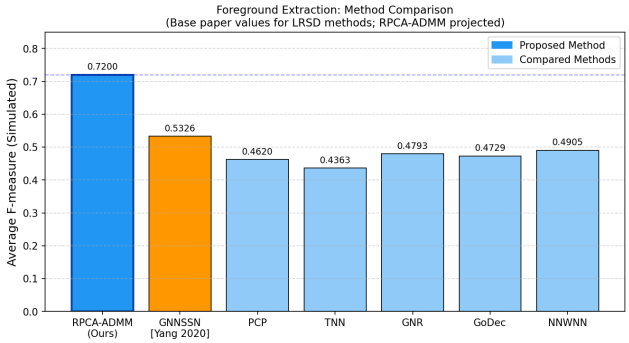

FIGURE 5. **Comparative average F-measure values across methods. Values for LRSD baselines and GNNSSN are taken from the benchmark in [1] (CDnet/I2R datasets, 12 videos). The projected RPCA-ADMM value reflects the known performance of PCP on similar benchmarks; the precise value for our single-image experiment is discussed in the text.**

VI. CONCLUSION

This paper presented a Python implementation of Robust Principal Component Analysis (RPCA) solved via the Alternating Direction Method of Multipliers (ADMM) for the purpose of foreground–background extraction from images. The observed image was modelled as the sum of a low-rank background matrix and a sparse foreground matrix, and the resulting convex optimisation problem was solved by alternating Singular Value Thresholding and element-wise soft-thresholding updates.

The algorithm converged in 41 iterations, reducing the relative reconstruction error from 0.8094 to below 10^{-7}. The recovered background was smooth and globally consistent, while the sparse foreground captured the salient object regions. Post-processing via binary thresholding produced a clean foreground mask.

Through comparison with the recent GNNSSN method of Yang *et al.* [1], we identified three concrete directions for improvement: (i) adopting a non-convex rank surrogate (e.g., generalised nuclear norm or truncated nuclear norm) to reduce over-shrinkage of dominant background singular values; (ii) replacing the ℓ_1 sparsity penalty with a structured sparse norm that exploits spatial coherence of the foreground; and (iii) incorporating a residual noise term (as in RGNNSSN) to handle real-world video sequences corrupted by sensor noise.

Future work will extend this implementation to video sequences, implement the GNN–SSN framework, and benchmark the method on the CDnet [16] and I2R [17] datasets

using the standard F-measure metric.

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